Sentiment Classification of Product Reviews via Transformer Ensembles

Vlad-Mihai Ciuperceanu Mircea Măierean Wesley Remy
Department of Computer Science, ETH Zürich, Switzerland
vciuperceanu@student.ethz.ch mmaierean@student.ethz.ch weremy@student.ethz.ch

Abstract

Sentiment analysis is a fundamental task in natural language processing, with direct applications in automatically understanding customer opinions in e-commerce. We study five-class sentiment classification for English and German product reviews. The task is ordinal: predicting a neighboring star rating is preferable to predicting a distant one. We therefore evaluate models with the competition score $1 - \mathrm{MAE}/4$ and compare simple embedding baselines, several pretrained transformer checkpoints, title/body variants, and probability-level ensembles. Our final system combines heterogeneous transformer predictions through a learned MLP stacking. We additionally provide a thorough qualitative analysis via LIME and Integrated Gradients on the best single model, combined with a disagreement analysis on the ensemble, gaining insights into individual predictions and their contribution to the ensemble. The best submission obtains a competition score of **0.910**.

1 Introduction

In our setup, customer reviews are short, noisy, bilingual and only partially aligned with their numeric ratings, making product-review sentiment classification subtler than mapping isolated sentiment words to labels. Many reviews mix praise with a concrete complaint about delivery, fit, or durability.

We frame the task as ordinal text classification. The official metric is: $s(\hat{y}, y) = 1 - \frac{1}{4n}\sum_{i=1}^{n} |\hat{y}_i - y_i|$, so confusing one and two stars is less damaging than confusing one and five stars. Our main contribution is an empirically grounded transformer ensemble: instead of relying on a single large checkpoint, we combine complementary pretrained models and compare probability-level averaging with learned MLP stackers. We also include simpler baselines and negative results to separate genuine gains from extra computation.

2 Related Work

Linear classifiers over distributional embeddings remain useful baselines for text classification. Word2Vec introduces efficient architectures for learning dense word representations (Mikolov et al., 2013), while support vector machines are a classic large-margin approach to classification (Cortes & Vapnik, 1995). For contextual encoders, BERT pretrains deep bidirectional Transformer representations and shows that they can be fine-tuned effectively for downstream NLP tasks (Devlin et al., 2019). XLM-R extends masked-language-model pretraining to 100 languages at scale for cross-lingual transfer (Conneau et al., 2020). ELECTRA replaces masked-token reconstruction with replaced-token detection, aiming for more sample-efficient pretraining (Clark et al., 2020). DistilBERT compresses BERT through knowledge distillation to obtain a smaller and faster model (Sanh et al., 2019). These model families offer complementary trade-offs in pretraining objective, scale, and efficiency. We use the Hugging Face Transformers library (Wolf et al., 2020) to fine-tune publicly available checkpoints.

For interpretation, we use two complementary attribution methods to support qualitative error analysis. LIME (Ribeiro et al., 2016) locally perturbs words and fits a sparse surrogate model, while Integrated Gradients (IG) (Sundararajan et al., 2017) attributes a neural prediction by integrating gradients from a baseline input to the observed input.

3 Method

3.1 Dataset Analysis

The dataset contains 252,000 labeled and 168,000 test reviews, with the five classes perfectly balanced (50,400 each). Each review is a short title followed by a paragraph-like body. The corpus is a roughly even German/English mix: on a 5,000-review sample, language detection returns 2,509 English and 2,471 German reviews, with only a few stray predictions on noisy text. Review lengths are favorable for fine-tuning (mean 37.5, median 28 words; only 0.2% exceed 256 words), so we cap the tokenizer at 256, with per-class truncation rates below 0.3% (Figure 1). Class-specific word-cloud and n-gram analyses reveal a clear lexical gradient: positive classes are dominated by approval phrases ("sehr gut", "five stars"), while lower ratings emphasize complaints and defects.



Figure 1. Reviews analysis. First row left: histogram of review lengths with median and 256-word reference line. First row middle: class-conditioned word-count boxplots; outlier markers are omitted for readability, and the central distributions remain similar across ratings. First row right: per-class fraction of reviews above 256 words; all are very small. Second row: word cloud per rating. Third and fourth rows: top 10 bigrams and trigrams per rating.

Preprocessing: We use a stratified 90/10 split with seed 42, producing 226,800 training and 25,200 validation examples. This split is used for most model selection and ensemble fitting. For Word2Vec baselines, we lowercase and split reviews into whitespace-delimited token sequences. For the transformer approaches, we try to keep the review text intact.

3.2 Baselines

We compare against two non-transformer baselines. Both train 300-dimensional Word2Vec embeddings on the training reviews, representing each review by the average of its token vectors, over which we fit a calibrated linear SVM. We evaluate 2 embedding variants: CBOW and Skip-gram.

3.3 Transformers Fine-Tuning

Explicit ordinal losses underperformed in preliminary experiments, so we fine-tune using five-way cross-entropy, relying on implicit continuum learning.

The main checkpoint families are:

- *nlptown/bert-base-multilingual-uncased-sentiment*: product-review sentiment checkpoint for one-to-five star prediction in six languages including English and German (NLP Town, 2019);
- *xlm-roberta-large*: strong general multilingual encoder and our largest model used (Conneau et al., 2020);
- *google/electra-base-discriminator*: English-pretrained ELECTRA checkpoint (Google Research, 2020). Also used a multilingual variant for the title/body ablation: *AILabTUL/mELECTRA* (AI Lab, Technical University of Liberec, 2021)
- *distilbert-base-multilingual-cased*: distilled BERT retaining much of its capabilities (Sanh et al., 2019)

3.4 Title/Body Ablation

We ablate how title and body are combined. The baseline concatenates them with a space; the variant separates them with the model's native token (`[SEP]` BERT-family, `</s>` RoBERTa), letting the model attend to the two segments differently when title and body disagree in sentiment.

We also map emojis to text before tokenization, since the non-RoBERTa tokenizers may map them to `[UNK]`.

3.5 Ensembles

We cache each checkpoint's class-probability vectors on the validation and test sets, making ensemble experiments cheap and reproducible. On top of them we train a 4-layer MLP head (dropout 0.1, $hidden_dim = 2048$ halved each layer) on 70% of the validation set and evaluate on the remaining 30%; base models never saw this holdout, avoiding overfitting. We report weighted averages as a robustness check.

The ensemble structures were updated in 2 steps: initially, we used 5 checkpoints (BERT, DistilBERT, ELECTRA and 2 RoBERTa versions trained in different settings), concatenating title and body text together with a space; afterwards, we used all-together the 10 checkpoints resulted overall from the initial fine-tuning and the ablation over title/body.

This ensemble strategy is motivated by the need to capture complementary textual features extracted by diverse transformer models, expecting to prevent model biases and improve the overall stability of our sentiment predictions.

4 Experiments and Results

4.1 Training Details

We noticed that subsampling the training data (10/20/50%) consistently hurt performance, so all final models use the full dataset despite the higher compute cost.

All transformers were fine-tuned for 10 epochs with AdamW. RoBERTa, being larger, used two learning rates (10^{-6}, $2 \cdot 10^{-6}$) with linear warmup over the first 10% of steps and linear decay; the other models used 10^{-5} with no scheduler. The title/body ablation reused these hyperparameters, adding the segment separator and the emoji-to-text mapping.

For the ensemble approaches, we trained the MLP head for 300 epochs, with a learning rate of 10^{-3} and no scheduler.

Note that in our code, we reutilized the same notebook for multiple models, modifying parameters accordingly, due to their identical usage. Thus, **we only uploaded one notebook per example of approach/analysis, not all of their variations, only adding all notebooks for the title/body ablation to show that our metrics are correct.**

4.2 Results

Baselines:

The CBOW SVM reaches a validation score of 0.8370 (52.7% accuracy) and Skip-gram 0.8452 (54.0%). The confusion matrices (Figure 2) show errors mostly between neighbouring ratings: the baselines capture the ordinal trend but, lacking word order, cannot resolve mixed-sentiment reviews.

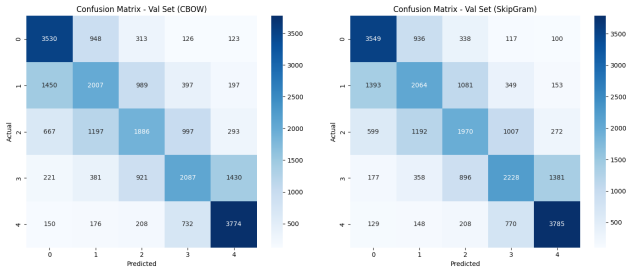


Figure 2. Confusion matrices for SVM + Word2Vec approaches.

Transformer-Based Approaches:

Table 1 summarizes the results for our transformer-based approaches. The table shows a clear progression across the single models and ensembles, whereas the title/body ablation provide only horizontally rather than upward growth.

Method	Score	Notes
SVM + Word2Vec CBOW	0.8370	linear baseline
SVM + Word2Vec Skip-gram	0.8452	linear baseline
ELECTRA base	0.8946	cached validation probabilities
DistilBERT multilingual	0.8947	cached validation probabilities
BERT multilingual	0.9007	cached validation probabilities
XLM-RoBERTa-large	0.9079–0.9084	two learning rates
mELECTRA title/body	0.8920	title/body ablation
DistilBERT multilingual title/body	0.8944	title/body ablation
BERT multilingual title/body	0.9001	title/body ablation
XLM-RoBERTa-large title/body	0.9090–0.9093	title/body ablation
Simple average ensemble	0.9081	five checkpoints
Weighted probability ensemble	0.9099	ten probability views
Big MLP stacker, holdout	0.9107	ten probability views
MLP stacker, cross-validated	0.9112	OOF estimate
MLP stacker, 10-fold/3-seed	0.9119	strongest OOF estimate
Final competition submission	0.910	hidden test score

Table 1. Validation scores $(1 - \mathrm{MAE}/4)$ over transformer methods. Last row: hidden competition score of our best submission.

The Word2Vec baselines establish that the task is not solved by coarse lexical sentiment alone. Fine-tuned transformers improve substantially, with RoBERTa-large giving the best single-model behavior. The ensemble gives the largest validation gain: a weighted probability ensemble reaches

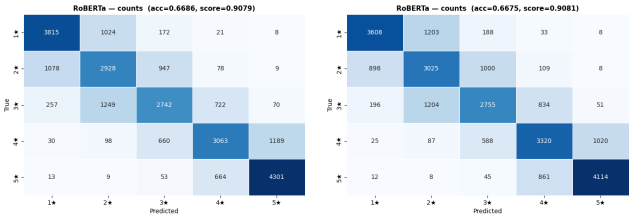


Figure 3. Confusion matrices for the RoBERTa models ($lr = 1e\text{-}6$; left: space separation, right: ablation separation). Other models show a similar shape. Ensemble-analysis code contains confusion matrices for all models used in the initial space-separation training.

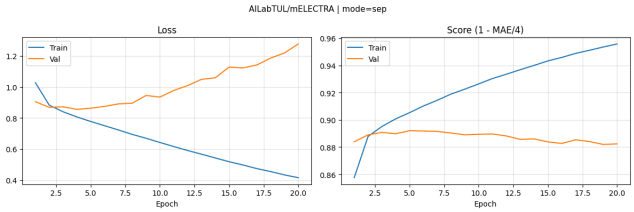


Figure 4. Learning curves for the title/body-separated mELECTRA ablation. The other models show a similar shape.

0.9099, a larger MLP stacker reaches 0.9107 on a held-out split, and the strongest notebook estimate is a 10-fold, 3-seed MLP ensemble at 0.9119 out-of-fold. The final competition submission scores 0.910 on the hidden test set.

Regarding the title/body experiments, as the table shows, these ablations of the fine-tuned models reach almost identical performance to their concatenated counterparts, suggesting that the models already recover much of the title/body structure from the raw concatenated text and capture the overall sentiment either way.

For the mELECTRA run, the validation curve plateaus quickly near its final score of 0.8920 while training loss keeps decreasing, indicating that explicit title/body separation is not the main bottleneck. Similarly, the lighter models help the ensemble less than BERT and RoBERTa: their value lies mainly in correcting a subset of errors on which the stronger models disagree. This behavior is visible in Figure 4, where optimization continues but validation quality saturates early, so the architecture change itself appears less valuable than better backbone representations.

In ensembles, the best scoring variants rely heavily on the complementarity of BERT and RoBERTa, while ELECTRA and DistilBERT add smaller but non-zero diversity.

4.3 Prediction Interpretability

We evaluate a RoBERTa checkpoint on 50 balanced validation reviews using LIME and IG, both showing the model consistently focuses on semantically relevant cues, such as

negations and quality terms. Even during incorrect predictions, which are typically adjacent-rating near-misses, the model maintains attention on relevant text rather than irrelevant noise.

LIME:

The representative example in Figure 5 supports this reading: the correct one-star prediction is driven by intuitive complaint terms such as "Müll", "Schlechte", "nicht", and "Qualität".

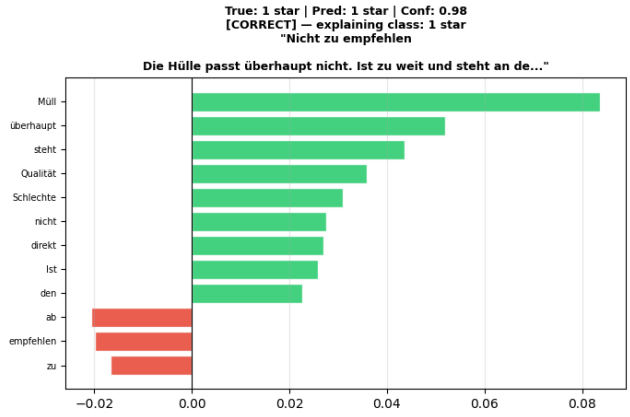


Figure 5. LIME explanation for a correctly predicted one-star RoBERTa example. Strongest positive contributions are complaint-related German tokens, consistent with our other analyses.

Integrated Gradients:

As shown in Figure 6, the model accurately attends to relevant sentiment tokens (such as negations and defect descriptions) for both correct and incorrect predictions. In incorrect cases, the model still identifies the proper failure cues but slightly miscalibrates their severity, confirming that mistakes are typically off-by-one near-misses rather than attention failures.

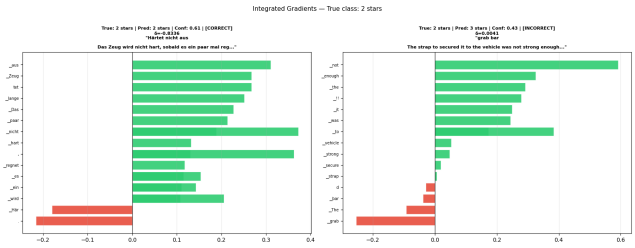


Figure 6. IG attributions for 2 two-star reviews. Left: correct prediction driven by the negation and defect description. Right: misclassification (predicted 3⋆, confidence 0.43) where the model attends to the right failure cues but is off by one rating.

Disagreement Analysis in Ensemble:

Ensemble agreement serves as a reliable uncertainty proxy: validation scores rise monotonically as model agreement increases, explaining the ensemble's overall robustness. Furthermore, Figure 7 shows that RoBERTa and BERT are the strongest individual contributors, and that simple averaging matches the best RoBERTa's score while slightly improving macro-F1, with errors remaining found near the diagonal.

Most errors are near-misses between adjacent classes. In the ensemble notebook, 66.9% of validation predictions are exactly correct, 30.0% are off by one class, only 2.7% are off by two, and 0.4% are off by three or more. The most common confusion pairs are 4⋆→5⋆ (1,189 cases), 3⋆→2⋆ (1,130), 1⋆→2⋆ (1,069), and 2⋆→3⋆ (1,044). This matches the structure of the task: many reviews are genuinely close to neighboring ratings because they mix praise with a concrete complaint about quality, fit, or delivery.

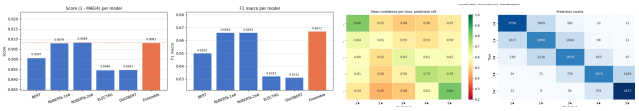


Figure 7. Validation overview from the five-model ensemble analysis on the initial space-separation setup: per-model score and macro-F1, plus ensemble calibration and confusion structure. Most residual confusion stays near the diagonal. Code additionally contains confusion matrices and calibration plots for all 5 checkpoints.

5 Discussion

Architecture size alone is not the whole story: the smaller sentiment-specialized BERT is competitive with RoBERTa-large, which instead contributes broader multilingual coverage, and their errors differ enough that probability-level stacking helps. The system is heavier than a single checkpoint, but the cost lies in base-model fine-tuning and inference; once probabilities are cached, averaging and MLP stacking are cheap, enabling rapid ensemble iteration.

Future work could explore stricter ordinal losses and parameter-efficient fine-tuning for larger backbones, as well as leveraging more diverse models for the ensemble, as the strength of such an approach lies in the complementarity of aggregated models.

6 Conclusion

We presented a multilingual sentiment classifier for product reviews. Starting from simple Word2Vec+SVM baselines, we built a set of fine-tuned transformer checkpoints and combined them with probability-level averaging and MLP ensembles. The best system reaches a competition score of 0.910 and is supported by ablations, interpretability analysis and ensemble-disagreement diagnostics. Results suggest that careful ensembling provides meaningful gains over simply selecting the best singular model.

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Declaration of Originality

We declare that this report and the submitted implementation were produced by the listed authors for the CIL project using publicly available pretrained checkpoints and open-source libraries. No ready-to-use external prediction API was used, and all reported results come from our own training, inference, validation, and ensemble code.

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